

# Roshan Hegde

 [github.com/RoshanAH](https://github.com/RoshanAH)  [roshanahegde@gmail.com](mailto:roshanahegde@gmail.com)  (952) 594-4175

## EDUCATION

### University of Illinois at Urbana-Champaign

May 2027

*Bachelor of Science in Computer Science, Grainger College of Engineering*

*GPA: 3.80/4.0*

*Bachelor of Science in Liberal Arts and Sciences in Mathematics*

**Courses:** Advanced Algorithms, Operating Systems, Programming Languages and Compilers, Abstract Algebra, Graph Theory, Cryptography, Data Structures

## EXPERIENCE

### W.W. Grainger | *Software Engineer Intern* | TypeScript, PostgreSQL, AWS

June – August 2026

- Engineered a test-code migration tool pairing deterministic code transformations and agentic editing to migrate Mocha to Vitest, reducing manual migration work by **80%**
- Integrated the migration tool across **12 repositories**, cutting CI/CD times by **64%**
- Designed a closed-loop prompt optimization system using automated benchmarking and feedback scoring, raising agent accuracy from **76% to 90%**

### Olam Group | *Software Engineer Intern* | Azure, Python

May – August 2025

- Analyzed utilization of virtual machines and compute instances using custom scripts and monitoring dashboards, cutting cloud spend by **\$30,000 annually**
- Built a self-directed data analysis tool to automate VM performance reporting, surfacing over-allocated compute resources for the infrastructure team
- Presented cost-optimization findings to the infrastructure team, informing their resource allocation strategy

### UIUC | *Operating Systems Course Assistant* | C, RISC-V, GDB

August 2025 – Present

- Developed programming assignments implementing core operating system components (concurrency, virtual memory, and filesystems) on RISC-V QEMU
- Rearchitected the autograder to grade student code concurrently, reducing grading time from **15 to 2 hours**
- Hosted office hours for **1000+** students, debugging student implementations of UART drivers, thread schedulers, virtual memory, filesystem drivers, and process isolation

## PROJECTS

### MechMania | *Lead Developer* | Go, Rust, Python, Nix

August 2024 – Present

- Replaced a WebSocket-based IPC protocol with a shared memory design using zero-copy serialization and exponential-backoff interprocess polling, resulting in a **1000x speedup** over the prior implementation
- Hardened security by using Nix and bubblewrap to sandbox execution of arbitrary participant-written code

### Operating System (LumonOS) | C, Unix, RISC-V, QEMU, GDB

January – August 2025

- Created a fully functional, RISC-V, Unix-based pre-emptive operating system with a round-robin scheduling policy
- Built a virtio block driver, cryptography driver, and entropy driver
- Implemented support for pipes between processes and argument passing according to POSIX standard
- Placed **3rd** in the Operating System design competition hosted by the ECE department

### GPT2 Inference Optimization Competition | *1st Place* | CUDA, C, PTX, Nsight Compute

December 2025

- Created custom CUDA kernels, profiled with NVIDIA Nsight Compute, achieving a **120x speedup** over the baseline and **2x speedup** over NVIDIA cuBLAS
- Implemented and tuned optimizations including flash attention, tensor cores, shared memory tiling, and fusion

## AWARDS

**1st Place, HackIllinois '24, '25** | Rust, Cargo

February 2024 and 2025

**1st Place, MechMania '24, '25** | Java, Python

September 2023 and 2024

## SKILLS

**Languages:** C, C++, Rust, Go, Java, Kotlin, Python, TypeScript

**Tools:** Git, Docker, Linux, Nix, Vim, AWS, Azure, PostgreSQL, GDB